



Similar triangles

TWO TRIANGLES THAT ARE EXACTLY THE SAME SHAPE BUT NOT THE SAME SIZE ARE CALLED SIMILAR TRIANGLES.

SEE ALSO

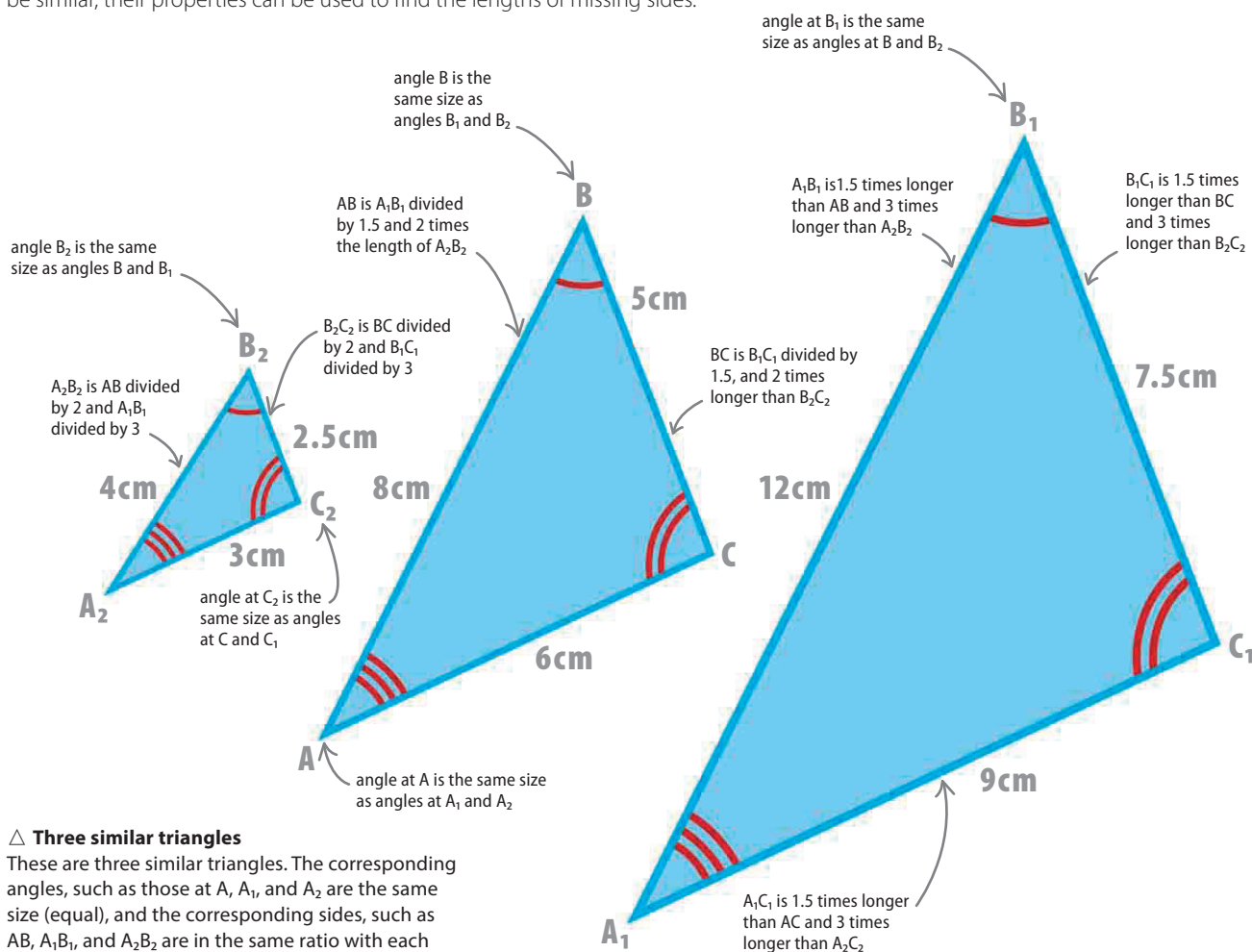
◀ 56–59 Ratio and proportion

◀ 104–105 Enlargements

◀ 116–117 Triangles

What are similar triangles?

Similar triangles are made by making bigger or smaller copies of a triangle—a transformation known as enlargement. Each of the triangles have equal corresponding angles, and corresponding sides that are in proportion to one another, for example each side of triangle ABC below is twice the length of each side on triangle $A_2B_2C_2$. There are four different ways to check if a pair of triangles are similar (see p.126), and if two triangles are known to be similar, their properties can be used to find the lengths of missing sides.



△ Three similar triangles

These are three similar triangles. The corresponding angles, such as those at A , A_1 , and A_2 are the same size (equal), and the corresponding sides, such as AB , A_1B_1 , and A_2B_2 are in the same ratio with each other as the other corresponding sides. It is possible to check this by dividing each side of one triangle by the corresponding side of another triangle – if the answers are all equal, the sides are in proportion to each other.